

# ActInf Livestream #033.0 ~ "Thinking like a State - Embodied intelligence in

*Livestream | 1:21:37*

me

[Music]

hello everyone welcome to actinlab live  
stream number

33.0 it's november 18th 2021

welcome to the actin lab everyone

we are a participatory online lab that

is communicating and mute if you're not

going to be speaking thank you we're a

participatory online lab that is

communicating learning and practicing

applied active inference you can find us

at the links here on this slide

this is a recorded and an archived live

stream so please provide us all with

feedback so that we can improve our work

all backgrounds and perspectives are

welcome here

we'll be following good video etiquette

for live streams and respecting our

awesome facilitator blue

at this short link you can find the

streams past and upcoming and we are

about to be heading into the second half

of november 2021 where on the 23rd and

the 30th we'll have two participatory

group discussions on this very paper

we're going to discuss today thinking

like a state

and then we have yet to set the december

papers so if you know someone who might

be interested in joining or you yourself

would like to discuss a paper of yours

get in touch with us

today in active stream 33.0 the goal is

to learn and discuss this paper thinking  
like a state  
embodied intelligence in the deep  
history of our collective minds by a  
guinin kalu  
and the video like all  
zero videos even though we do have the  
author here it's just an introduction to  
some of the ideas  
and how to pronounce them not a review  
or a final word so we'll introduce  
ourselves and then go into aims and  
claims abstract and roadmap  
keywords figures and formalisms etc so  
let us know especially if you'd like to  
participate in the upcoming 33.1 and  
33.2  
we'll start with just an introduction of  
saying hello and what we're excited to  
talk about or resolve either in the dot  
zero or in the upcoming group  
discussions  
and then we'll culminate with the author  
and transfer facilitation over  
so  
i'm daniel i'm a researcher in  
california  
and  
i'm just excited to  
check in on the progress of this  
important topic people have been  
theorizing about societies and cities  
and states and everything in between  
since as long as they were there  
probably a little bit before  
so it's just a fun topic and i'm really  
excited to see how active inference can  
give a new lens at resolving some of the

long-standing questions in that area

i'll pass it to stephen

yeah thank you daniel um

i'm steve and i'm based in toronto and

uh

i'm really interested in the multi-scale

implications of

state and

niche construction particularly for me

because i'm interested in working around

what i call social topographies which is

probably at the scale of

events and moments in people's like

lived experience and in their

architectures and places so i i'm really

excited that this is bringing in this

idea of state

and larger scale but also bridging that

with

active inference from the kind of

faster smaller scale so yeah lots of

good stuff here lots of ontological

questions as well which i think is

really cool

um and uh i passed it over should i pass

it over to blue blue or sharp yep over

to you

so i'm blue i'm an independent research

consultant in new mexico and i really

like this paper

and i really like working and talking

with avell about ideas because these

ideas are kind of at a bigger scale

i am interested in self-assembly of

things like at the cellular and maybe

even subcellular level and like this

takes it to the very big level and i

think that active inference really kind

of unifies this it's a skill-free  
framework and so that's kind of what i'm  
excited to talk about today and maybe  
not resolve but discuss and get some  
other perspectives on um and i'll pass  
it to the first author of the paper or  
the only author of the paper avail if  
you'd like to just introduce yourself  
say a few words

so i'm  
available

i have a training in collective science  
and physics mainly i'm interested in  
philosophy and a lot of things

what we learned here is that i funded  
and helped organize cairo's research  
participative citizens laboratory which  
aim is to understand

how to act or to navigate

contemporaries and show historical  
dynamics

and part of this is to understand

why we have states and

how those states that we now have are  
like

um

so the goal of the paper is to basically  
understand how states have constituted  
and what it tells about their continued  
structural identity through ages and  
this

of course the topic is big there are

several kind of states but the very

first thing we need to do is to

understand what your state is

and where it comes from and that's what

i'm trying to do there

awesome so moving on to the aims and

claims of the paper  
uh so this paper thinking like a state  
embodied intelligence in the deep  
history of our collective mind  
in the paper it states that the core  
argument of this article is that states  
can be and should be understood as  
hierarchical control systems essentially  
similar in their core physical  
architecture to brains  
and we will leverage active inference to  
ground the study of intelligence in the  
way an organism maintains their  
structural autonomy to formally assess  
the collective brain hypothesis  
and the big question of the article  
is what kind of intelligence is  
displayed by city-state systems  
and how is it embodied in their  
structure

so we'll go through the abstract daniel  
do you want to take the first section of  
the abstract

yeah stephen first

oh just one comment just because i know  
you mentioned kairos there and  
i've done an activity with the kairos  
blanket

or it's like a mapping  
thing where you stand on parts of canada  
and to look where it came from so i  
don't that connects to i if i'm writing  
you kairos is  
working at that kind of regional  
national scale to look at  
questions around that and  
from what i hear that's informing  
how your research is

being approached is that correct  
i did not understand the question um  
not correct  
yeah it was an activity it's an activity  
actually not to do with a fristen  
blanket literally where they put a big  
blanket um about  
parts of canada on the floor in a room  
and then people have to stand on it and  
they keep taking away bits  
which were basically um  
which which were not are no longer  
in the territories of the first nations  
and then you see how much is left  
there's not a lot left  
so to speak so it's talking to that idea  
of statehood and  
settlers and how much is left and i  
wonder if that speaks to the wider work  
of kairos and  
questions of states and how do we  
understand how we fit into that or maybe  
i'm i'm mis  
misunderstanding  
so that fits in the work of keros  
because it uh tries to bridge scales  
i never heard of these keras broadcast  
this is not related to what we do in any  
environment it helps embody an  
impression of how well settlers have  
appropriated  
indigenous american territories so this  
is something that's  
called that fits the mandate of karus  
but the core  
idea of keroses  
it's the greek notion uh for time for  
three in a greek and carlos is the

instant uh and it's uh coded  
actively like it's an instant where you  
can see the opportunity  
and um this is basically what embodied  
agents do they build skills and they  
build capabilities and they enact them  
when it separates this is the goal of  
chaos to understand and reproduce uh  
this uh core ability and  
a lot of people have used kerosene you  
have anarchists  
that claim um that are called cairos you  
have a christian concept that is called  
cairos you have a lot of people that  
make the use of the concept because it's  
very broad and it's very um  
it's not something that has been  
imported in the  
christian concept of times really in  
like what you have today so if you want  
to talk about this you use you just use  
the greek word which is kairos so  
yep just one thought then i'll go to the  
abstract which is that kairos as opposed  
to kronos like a chronometer with  
decimal time kairos is like timeliness  
and it brings action and sort of  
a little bit of a moment that transcends  
the tick tick tick tick tick so it's a  
cool term  
i'll read the first  
third of the abstract  
this article aims to show how the deep  
history of early state societies entails  
the development of a collective form of  
cognitive agency  
it relates classical works in the  
anthropology of states

in particular scots seeing like a state  
with the inactive account of biological  
and cognitive organization thanks to the  
unified ontology for self-organization  
dynamics across scales offered by the  
active inference framework

stephen do you want to take the next  
piece of the abstract

yes

okay

so active inference conceives of  
cognition

a synchronization across individuated  
sensory motor states

it entails that biological or  
sociocultural constraints display  
minimal form of cognition by shaping the  
behavior of faster dynamics in a certain  
way

when such constraints collectively  
define a basic life form

an integrated operationally closed  
system

they can therefore be said to embody  
adaptive knowledge properly speaking

and then i'll just take the last part it

says the inactive inference account i

articulate here strongly motivates and

methodologically grounds a holist

a holist approach in the social sciences

indeed it grounds the study of human

societies in the role of structural

constraints whose meaning depends both

on the broader systems activity and in

the historical context of their

emergence

the present account of the dynamics of

early urbanization and state genesis



aims to illustrate this approach  
and a quick just walk through the road  
map of the paper we're gonna go through  
the introduction section and then there  
are three main sections of the article  
the embodied intelligence in nested  
minds is the first  
the second is the deep history of cities  
and states  
the third from the urban metabolism to  
the collective mind and then  
we'll go through the conclusion  
and first up in this  
dot zero contextualization video we're  
going to discuss the keywords which i  
think there were no official keywords  
but these are the words that i created  
for the paper when discussing it  
at the conference so we pulled out  
intelligence  
autonomy embody intelligence active  
inference the free energy principle the  
concept of thinking through other minds  
and we've done that in a stream before  
daniel is that true i think we have  
right  
um and then uh niche construction so i  
don't know i don't have the facts on the  
on the exact number of the thinking  
through other minds um stream but we can  
find that out somewhere during this  
during this talk we'll look it up  
um  
so  
even do you want to maybe speak to we'll  
give you the first stab at talking about  
what you mean by intelligence and  
autonomy we've pulled a couple quotes

out of the paper but maybe you'd like to  
say a few words  
so if i remember well intelligent  
etymologically it's the capacity to make  
links  
to connect dots of data and to build  
antiquity understanding of it  
i don't want to commit to any specific  
notion of intelligence because  
i am not aware of any meaningful  
first principle definition of it  
there is a lot of work in measuring  
intelligence like iq  
which  
maps to specific cognitive abilities and  
has literally zero reason to generalize  
to other life forms than humans  
and we contribute to other life forms  
than humans in occidental societies  
the national autonomy is much more  
specific it's important to uh  
okay so etymological is one's ability to  
build one's own rules so for example if  
your city state has laws and  
the organs that make the law are within  
these states it is autonomous in a  
strictest sense  
it is also recruited in the study of  
life and condition and the way cognition  
is grounded in life  
by inactivist scholars  
where it is  
related to a personal closure under  
precarious consensus  
so it is something that is  
able to operate and able to recreate the  
conditions in which in which it operates  
while there is no uh exogenous force

that help himself do so

for example if you have a candle it

helps

there is a burning fire which uh

generates a flow from season so as to

maintain itself but if someone does not

keep um

fuel going

this dies so this has no really way to

recreate the conditions at which it

operates this is in contrast to the

simplest bacteria there is which is able

to look for food basically

so the

active concept seeks to capture that

nice um daniel do you maybe want to read

the the quotes from the paper directly

referring to

um

intelligence and autonomy as described

er intelligence is the general ability

for a system to understand the world

either in a reflexive propositional way

or as enacted in their ecological

activity

and autonomy refers to the property of

operational closure under precarious

circumstance ie to the capability of

biological systems to continuously

maintain and recreate their own

structural identity as a result of the

collective activity of their

constitutive processes

right and these are processes that are

actively generating and sustaining

identity

and i think avell spoke really well to

operational closure um in the what you

just described but we just want to refer  
anybody who's listening to  
this paper by  
monteville and massio in 2015 this  
biological organization as a closure of  
constraints as closure of constraints  
so i want to emphasize  
that there is i think it's pretty old  
the percentage it has at least 10 years  
and the monteville paper is an attempt  
to formalize it in uh logics  
so it is not necessarily the the  
reference on this i don't remember what  
is the reference paper on this

[Music]

awesome thank you  
um maybe let's move on to embodied  
intelligence and stephen if you uh this  
is perfect for you if you want to read  
the quote from the paper describing um  
what embodied intelligence refers to  
yes okay  
embodied intelligence refers to the  
ability for understanding  
adaptation so sort of a mixture of both  
that is imprinted in an agent's physical  
structure rather than in some  
general computational ability  
yeah that's significant and then we have  
the wood lice  
alias yeah don't use the jitsi chat fyi  
you can use key base if you want a back  
channel  
but that's the wood lice example that  
carl firsten provided in the 2018  
example that's a little shout out there  
awesome  
um

yeah so

the uh does anybody want to say a few  
more words about embodied intelligence  
maybe avel or steven or if there's  
something that comes up for you guys  
feel free to raise your hand

if there's something else you want to  
discuss

and the slides are the slides that we  
prepared avail for the  
talk they're in the key based chat as  
well

so i'll i'll mention just to say i mean  
because we're kind of introducing at  
this stage i won't start raising those  
and loads of questions and  
we'll save that for the the next one but  
uh and i just note that um  
that this is useful just to highlight  
how this is being structured because  
you've got this word intelligence  
embodiment and

uh i know in the there's them talk about  
the brain or the idea of the brain and  
the idea of the mind

um so i think it's interesting here that  
um

this this paper sort of bridges a bit  
across

um the brain

intelligence mind culture kind of

uh area so when it's talking here it's  
kind of seems quite an open description  
um

and is um

really making sure it's clear to  
speak to the difference to this general  
computational ability

and that it's it's somehow a different  
type of  
uh knowing that's been done maybe with  
the word you know morphological  
computing sometimes is used in that in  
that world  
nice thank you  
um maybe daniel do you want to  
explain the free energy principle and  
active inference  
sure so  
as on the slide here the free energy  
principle states that dynamical systems  
given sufficiently regular boundary  
conditions which are required for their  
continued existence  
spontaneously self-organize so as to  
minimize their expected variational free  
energy through time  
effectively maximizing evidence for the  
world model they enact  
so the expected variational free energy  
which we talk about in more detail  
elsewhere it's kind of like all aspects  
of the system through time are what have  
to  
be handled in order for persistence  
and  
using some technical formalisms  
that turns out to have  
equivalence with evidence for a  
generative model where one is persisting  
active inference explicitly brings  
action into that inference process and  
it's the process by which dynamical  
systems maximize their evidence for the  
model of the world which they enact this  
can occur by a system altering its

environment to fit expectations i.e  
niche modification  
so it's the difference between  
the free energy principle merely saying  
that persistent systems are going to be  
isomorphic with those that have high  
evidence for the world model which they  
enact and then active inference as a  
corollary process theory says  
and the way that they're going to do  
that is through  
inference updating their internal model  
or through action  
which can be  
just sort of like a flock of birds  
action in the sky or it can be niche  
modification like stigma g  
nice thank you  
um and then we can go on to thinking oh  
stephen you have a comment to say  
yeah just one piece here also sort of  
flags up for me is the the use of the  
word model because we talk about their  
own model of the world and yet um  
often we think of models as something  
that we've created in the world you know  
on a piece of paper and things like that  
so again there's a sort of ontological  
challenge in the sense that  
it's it is kind of like a self model but  
it's  
it's it's a self model that's not a  
representation  
um so um but our language is kind of  
couched in our history  
um so anyway i think that might be  
something that sort of resurfaces and  
have to kind of

keep restating because people might have  
this representational  
kind of inbuilt  
way of  
you know those metaphors are quite  
embedded so it's hard for people to  
necessarily switch to what it means to  
move away from representation  
representational approaches  
i have one question as well  
the reason why we're talking about  
active inference and free energy  
principle in this paper so  
avo  
what  
does  
active do for this paper are there other  
contender theories in the game how else  
are people trying to take a unified  
approach to studying  
the specifics and the generalities of  
states  
so the um  
the approach i have i define as an  
active inference with the n  
between parenthesis it's a synthesis  
between what is said by an activist and  
what is said back to influence theorists  
the core framework is inactive i'm  
interested in um  
characterizing states uh  
state societies  
as a network of constraints so as uh  
slow processes  
that are  
exist at the holy scale  
uh not with the individual not between  
individual but in some sense over



individual

and that

shapes the activity of individuals so as

to maintain itself that's the core

argument society is our organism but now

if i want to get there i need uh some

strong instincts

i if i need to get there in if i need to

formalize the role of states which are

at core information processing machines

in it i need to formalize in a struggle

in a stronger sense the relation between

um

public synchronization basic

authorization and epistemic states and

that is precisely what active influence

does

so i need uh the two

uh i need elective uh

formalism the native approach and i need

active influence formalism to

make the link between the

ecological activity of states

as an international process and the

body the

constraints that actually consists

that actually statistics consist in

[Music]

nice thank you

we can i guess move on to

uh thinking through other minds

um stephen did you have a comment just

one other comment as i was interested by

the um you know you're using the term

constraints

so

and in active inference is sometimes

thought of as the higher level processes

are in some ways constraining the  
behavior of lower level processes rather  
than necessarily dictating them so  
they're you know it's look like i'm  
doing the faster jig  
and someone's just  
slowing me down a little bit um  
and so on so  
yeah just curious that that word  
constraints has brought in because often  
people think of  
you know intelligence is all about doing  
things and  
you know not constraining things so  
anyway just curious about that  
yep that's related to synergetics hawken  
not bucky and about like a wave and a  
ripple like the wave constrains where  
the ripple can go the faster  
fluctuations are constrained by the  
slower ordering parameters  
so here we have thinking through other  
minds  
and this  
is another body of work which we won't  
spend a ton of time on but  
for  
organisms that are making generative  
models of their niche  
sometimes that niche contains other  
organisms such as their self  
and it matters how those other organisms  
also include generative models of  
oneself so thinking through other minds  
is describing how in a shared epistemic  
niche shared informational niche with  
shared salience cues for example  
that two different agents are able to

coordinate

because they are thinking through each other's minds like if i put up my hand then blue will think that i want to speak

so it's sort of like a little bit of a theory of mind but with more of a act imp flavor

so both like the idea of constraint and this concept of thinking through other minds bring me back to mike levin's paper the computational boundary of the self because of of the nested like cognitive cones right and when he kind of talked about it he talked about you know the the higher order or like the larger level the higher level organism um or collective constrains the behavior of like the the individual um like sub level collectives um so that was kind of this idea of constraint and and then also um like thinking through other minds like specifically we talked about um

how

connections between cells individuals but the connections that that essentially breed information sharing so so when you have like a shared informational niche that enables this this process kind of of thinking through other minds where you think more as a collective and so that expands the cognitive boundary of the self to then include other you know individuals within that cognitive boundary stephen

yeah and that ties into the work of andy

clark and uh the extended mind so  
there's a  
um so the way that that gets uh there's  
there's a bit of um talk about how do  
you think about that extension happening  
and how much is it an extension of the  
individual and how much i suppose is  
what's being talked about here is how  
much does  
what has been extended start to become  
its own  
sort of cognitive agent  
nice and maybe very appropriately avail  
has returned and we are moving on to  
talk about  
niche construction and i think um avel  
might have  
a lot to say about the niche and its  
role in  
extending the cognitive boundary um so  
maybe do you want to give us a few a few  
insights into can we help us think  
through your mind  
modify the audio visual i'm not sure  
what i could say about the niche  
sorry modify the audio visual niche of  
our joint  
well that's attention  
we modified the niche i will actually  
actually within english i don't know  
anyway um  
i can't change your model of the niche  
this is not what i'm going to do  
i think  
i'm not trying to do that activity at  
least uh whatever um  
so the there is body of work in  
biology

uh evolutionary biology  
which is centered on how organisms  
modify  
their direct environment  
in a way that somehow fits their  
their ecological needs  
so a typical example is how beavers will  
be  
dumb source to get i i honestly don't  
know what beavers get out of them but  
they get something  
and  
the basic arguments uh in the um  
thinking through of the minds uh  
framework is that the way humanifier  
niche has a cognitive value it's  
externalized expectations  
and  
by analyzing expectations it allows  
expectations to synchronize between  
individuals  
and  
it's  
there is a very clear  
analogy or identity between  
once you accept there is a cognitive  
value in a niche construction as the  
externalization of expectations  
there is a very clear analogy and even  
identity between uh niche construction  
and excellent cognitions beaver are just  
extending uh their cognition  
their expectations the domain of the  
world that  
somehow participates  
to the fact that they um  
they live they keep on leaving and  
understanding the world

so

there is a pretty basic way in which an  
organism can expand  
the

extent of the world the domain  
which is operationally closed and which  
participates actively in their  
maintenance

and

this is basically

what's

there is to be found in this  
construction by shaping their niche  
organism shapes

their world they shape

their cognition they shape

their body

and

i'm not sure

i'm trying to give a general insight in  
uh initial section and how it relates  
but i'm not sure

how

relevant it is because i'm not sure what  
is the point that is to be made

followingly

just that niche construction is an  
important mechanism of

extended cognition like the recent  
stigmargy models show that individuals  
don't need to carry on board or

internally the entire cognitive  
apparatus they only need to know how to

interact and potentially modify a niche

and then also i think this is important

when we're talking about

cities and so on and states because

our niche can as per thinking through

other minds include other  
others like us  
or distant from us but also it includes  
the abiotic niche so climate change and  
resource utilization those are niche  
modifications there are some radioactive  
mountains  
that have been niche modified and so it  
it provides a continuity  
between  
the kinds of ecological changes that  
humans make  
with the cognitive approach to  
ecosystems and so that's why it's a very  
rich framework to build off  
so i just want to read this quote from  
the paper really quick and then and then  
stephen will come to your comment so it  
says as stated previously the network of  
constraints defining the structural  
identity of basic life needs not be  
printed in flesh it only needs to  
influence faster processes so as to  
continually recreate itself  
and there the references for this are  
there on the slides if you care to probe  
further it says such constraints can be  
embedded in the informationally rich  
sociocultural niche which enable the  
human ecology by affording adaptive  
coordination and the development of  
complex skills and i i think one of the  
the main things that um this paper does  
and uh feel free to correct me if i'm  
wrong but it kind of takes the idea of  
cumulative cultural evolution  
and  
brings the niche into this as a

cognitive extension

so i think that's one of the key moves  
made here

basically i know a cultural revolution

uh i have known revolution for like five

years or so actually in france for one

year only so cultural relationships so i

see things and i'm currently writing a

paper on uh that could be integrated on

reductionism in cultural

and the fact is cumulative cultural

evolution in cultural revolution

is the replacement of low fitness

variants by high fitness variants

within a variant i mean of cultural

tokens that are implemented within minds

our behavior because there is not a

strong ontology on what the replicator

is in cultural evolution

and

a basic inside

world to convey is that

culture does not stop in the brain

culture does not certainly does not stop

in uh individuated cultural cocaines

and

one of the goal of the paper is to show

that social revolution it has analyst

extension uh our intention i don't know

uh

it does it exists uh not only within the

minds of individual and within uh the

verbal effects of cultural tokens it

exists

[Music]

instructors that are not apprehensible

uh by an individual

and that are not entailed by any



specific cultural token

hence the holism

thanks thank you stephen

[Music]

yeah also bringing that beaver example

is

to some extent the beaver constructs a  
dam and to some extent maybe it grows a  
dam

um maybe in the same you know so by  
adding pieces it's it's growing

something a bit um and at what point  
does it

sit stand back and say oh that  
move that one a bit to the left move  
that a little bit to the right and sort  
of

improve the construction and how much is  
it an evolving process

i suppose there's also the idea that  
once something has grown  
and it's changed the niche  
um something else

you know just like the brain effectively  
has to grow before it can do its  
thinking um so maybe there's something  
about

how these structures have grown  
and been constructed so i don't know i  
won't i'm just mentioning that

awesome thank you um i guess uh maybe we  
kind of got ahead of our of ourselves uh  
talking about niche modification and  
cultural evolution um i love this

picture because it really uh shows like  
kind of um how much we have offloaded  
our our cognitive selves into the niche  
environment like we don't need to know

how to use directions or what  
restaurants are good or we don't need to  
know actually any of that anymore  
because it's all like riding around with  
us in a smart phone in our pocket all  
the time um so so we've just put put our  
brain we've offloaded all of the  
cognitive capacity of our of our brain  
or at least a good portion of it so what  
do we think about now what makes me  
think about that always um it says in  
the paper we claim that niche  
modification is critical  
for cumulative cultural evolution and  
advancements such as writing allow  
active inference to occur between agents  
and stigmatic cues embedded in the niche  
although  
did i say what you did i write what you  
just said  
okay  
what i want you to note is that what you  
described the way we offload the  
information on our social analytical and  
technological  
environments is the classical uh  
collective brain picture which is  
entailed by uh  
mostly um void which has shown in  
handwritten of cumulative cultural  
evolution  
uh what i want to show is that uh  
together step beyond  
we do not only have a self that is  
augmented by  
environmental cues there is a self  
outside ourselves there is uh things  
that shape social life and that are not

their agency that shape social life and  
that is not human that is institutional  
so that's the step beyond the collective  
brain in the vanilla a collective brain  
view which is maybe an anti-collective  
brain there is  
actually such a thing as a connective  
brain starts an image and we can study  
it  
well we can study it now through  
wikipedia and the internet like our  
network of the collective brain steven  
yeah it's interesting as well in terms  
of  
we have all this extended information  
but  
there's this there's a very good book  
called the paradox of choice  
so the the thing is we have all this  
information but you still have to do an  
action policy  
on what to do  
and the challenge is you can have so  
many you know it's like going to the  
supermarket and you know you can buy  
50 different types of olive oil  
okay it's just olive oil at the end of  
the day but now you've got to choose  
between 50 and you've got you know so  
this there's a sense of abundance but  
there's also a need for an action policy  
which is of course what active influence  
talks about um that ultimately it's not  
the volume of information that's going  
to be  
the game it's the ability to  
act  
and uh inappropriate ways um and make

sense of how to be in your niche so  
that's kind of interesting  
awesome thank you  
um so we are gonna move on um we're  
gonna go into the next section embodied  
intelligence in nested minds but but  
just to quickly  
summarize here like what was the main  
the brought up in the introduction so  
we've gone through the keywords but just  
to clearly just state it out autonomous  
systems are defined by a set of  
processes that create and sustain their  
identity they embody their understanding  
and adaptive abilities within their  
physical structure they enact a model of  
their world and seek to maximize  
evidence for this model through active  
inference  
and they can maximize model evidence  
through modification of their niche so  
just to kind of sum up what an  
autonomous system is and what where we  
are at this point and then we'll move on  
to um  
discussing embodied intelligence in  
nested minds  
so a quote from the paper it says uh  
here we argue that innovations are  
instead an emergent property of our  
species cultural learning abilities  
applied within our societies and social  
networks our societies and social  
networks act as collective brains we  
outline how many human brains how many  
human brains which evolved primarily for  
the acquisition of culture together  
beget a collective brain oh this isn't

actually from from the thinking like a  
state paper this is from the innovation  
in the collective brain paper  
and this framework this collective brain  
framework grounds the study of  
intelligence in the way that an organism  
maintains structural autonomy and allows  
the formal assessment of the collective  
brain hypothesis so the collective brain  
hypothesis has been  
put forward but not formally assessed up  
until this point avail do you want to  
talk a little bit more about that  
because i probably just botched it  
about how the collective brain pathogens  
is not commonly assessed  
yeah

so  
in cultural evolution you have  
what i'd say is the dominant theory is  
gen coevolution uh gene culture  
qualification theory  
which is basically centered on how um  
we have cultural variants that have an  
adaptive role and there are a set of  
cognitive bias  
or cognitive mechanism but actually  
cognitive bias uh bias actually actually  
that help us identify what are the most  
adaptive cultural variants and this uh  
has a feedback effect on gene selection  
which has an effect on cultural  
selection and  
basically all of your interpretation is  
until the process of  
a convolutional process between a  
genetic and cultural  
evolutionary information

processes

and

the fact is

there are claims there are numerous

claims in the literature that

there is some sort of collective

computation that you entered our

corrective cognition that is entailed by

this so there is if i remember well uh

henry and mata krishna who wrote the um

innovation in the collective brain paper

and there is i think boyd and um

svaldino

boyd which is

basically one of the two founders of the

initial formalism and smart dino which

is

one of the most competitively study a

member of the current literature that

wrote a paper on a collective cognition

no

collective comp distributed computation

because revolution has discrete

computation i don't know neglect data

but it's

uh i want to accentuate it a lot it's

words like they have a model of

individual bias

in a selection

and they have a narrative interpretation

of this somehow in telling some kind of

computation and there is

nothing in between there is a few paper

of

less introductors that's kind of

interested in the convolution of

social networks and cultural variants

access and smaller to name them

there is

literally not one paper that explains

how is this supposed to be cognition

what are the mechanisms what is the

cognitive architecture there is nothing

on it

uh or it's marginal and i did not heard

of it

and

the goal of the paper uh

is to basically explain how we can grant

a view of connective mechanisms that are

entitled by the social cultural networks

and structural constraints that are

embedded within a society

and that's it's not a model because

uh to build a full causal model of any

brain or organism is not

tractable but it's to prove that it's uh

meaningful

it's a

there is a positive way in which we can

model societies as or at least process

within societies as a collective form of

cognition

and a key example of this is the state

because we have access to the mechanisms

we have access to different papers with

which they understand and predict and

control reality

[Music]

awesome that's a perfect i think

transition into the next slide

do you want to read this first daniel

this paper uses the framework described

to illustrate that the co-evolution of

top-down control and cultural niche

construction enabled the inversions of

the city-state complex and how these  
collective organizations maintain  
structural autonomy  
awesome thank you  
and so that's kind of the  
the summary i say of the of the section  
the embodied intelligence and nested  
minds uh section does anybody want to  
give a thought or we can go on to the  
next section  
just what you just said about how we  
know the mechanism level of the state  
um  
i'm thinking  
we see its neurotransmitters and its  
action selection  
policies  
so it's like  
every system has advantages and  
disadvantages for its study  
this is one where many of the key  
features  
are designed and so it's a lot more like  
investigating the connections of a  
program that was written  
rather than the potentially like map  
territory debates over like what is the  
bacteria really doing are we really  
measuring the bacteria but it's like we  
know how certain processes in the state  
happen  
but not all of them and citizens usually  
don't know  
steven  
yeah so following on to what daniel said  
a quick question in a way is  
how  
much you



may be trying to have a sort of a  
realist approach on the idea of the  
brain or being a  
brain or how  
how much is it an instrumental tool or a  
metaphor so be curious as  
you know you take that correlation  
because you've got brain you've got mind  
you've got intelligence you've got  
extended cognition  
is it that you are really wanting to  
make it say that is like a brain that's  
kind of  
in  
enacting its own  
action policies or  
um so that would be that might be  
helpful  
so i think actually like maybe we should  
hold that to the end section from the  
from the because i think we're going to  
get more into how is a city like a brain  
um as we get closer to the end of this  
contextualization we'll save it for dot  
one and dot two because we'll cover just  
the background here but it's a very good  
question  
yeah very good um it's going to provoke  
a lot of discussion so that's why i'm  
like well maybe we should get through  
the rest of the rest of the  
talk going on first  
so um  
it is  
so we're moving on now to the deep  
history of cities and states  
uh here is another quote from the paper  
uh stephen do you want to read this for

us

yes okay

cultural niche construction

allowed humans to alter their social

cognitive ecology

in a way that allowed extensive

functional specialization and the

evolution of supra

individual agency through mediating

institutions

that makes me think about the evolution

of functional specialization in ant

colonies

and it's like

you can't have the super specialized

task you know you can't have somebody

who specializes on just sourdough bread

until there's somebody else who can help

them with some other tasks

and that's the economic inspiration for

why they call that process in ant's

division of labor

downstream of adam smith 1776 the

economic division of labor and the

candidate maker and the shoemaker and

all of that so once you have a lot of

people living together they can

specialize in a new way there are new

institutions needed to mediate

and that results in more being different

i love how daniel like says ants and

gets like giddy and excited

a stephen

yeah and i noticed this term super

individual so there's kind of this

um i suppose in the anchor you think of

the queen but in in our world we we have

these

and actually i've been thinking about a lot of this myself in terms of how we relate to these bigger scales so this idea of a supra individual or supra collective ties into the idea of the state you know because you know we think of you know the word state is actually used literally in the united states to be you know a region but there is an idea that is trying to maintain something so um that term i think is a is a useful one to sort of put out and uh and that is super individual agency so that ties it into active inference in a in an interesting way um it would be interesting to know the term socio-cognitive ecology it's not one i've heard before in quite that way so i don't know that could be something just to just to maybe ground that term in this context well we can return to it and i added the book cover for leviathan with the person made up of other people awesome thank you uh so um in this section of the paper avelle explains how state niche niche construction depended on serial monoculture uh the technology of writing material infrastructure uh dense

sentiment dense settlements  
and engineering societies from the top  
down as documented in seeing like a  
skate seeing like a state by scott  
um avail do you have a few maybe words  
to say about this i mean more than the  
brief kind of overview i just gave  
so i'm not sure identified a key uh  
subject like on the um  
importance on the uh morphogenesis of uh  
states from a social scientific  
perspective  
just we're just talking about state  
niche construction like what what did  
that depend on  
all of the things  
um  
but the career states is um  
i think we could have a lot of  
definitions uh their campaigns it i  
think it's okay it is said in certain  
tenses that it is uh  
essentially a monopoly of on legitimate  
violence  
so if there is a group of men  
that has weapons and that has the  
exclusive rights to use it and that this  
right is  
probably accepted it is a state  
uh i'm mostly interested in the mode of  
operation of a state so for me it's a  
chain of command basically it's an  
emissive systems that  
makes people do things  
so these those two definitions they  
entail  
the loss of human autonomy then tell  
that you have at least a subset of

humans that  
are constrained to follow the laws laws  
that are uh extremely imposed  
so that's in clear deviation from what  
is done say in iroquois society where  
at least by the personal contact  
mostly no one has the power to coerce  
anyone else and any decision is grounded  
on rational debates  
so this is not what a state is like a  
state is when you are a guy with the  
biggest bird and you can say okay you go  
kill this person because i want to i  
want this weapon and

um

you have a

[Music]

definition of the states that's

uh sorry

your definition of the state there is

that is uh grounded in the

sub-brain i'm not sure when it initiates

i think it's with leviton which i did

not read but basically that the state is

a central actor that has power um on

anyone

which is um sorry of

chats

things

um

so state is uh something that has power

on anyone it's not tied to a specific

individual

so that entails the capability for a set

of for institution or a set of

individuals to

basically coerce other people to do

things

and this entail  
for example enough at least a north  
centralization of resources for other  
people to not flee  
or enough imbalance in wealth for other  
people to get to you even though they  
know it will strip them of their  
autonomy so it's about the grounding on  
which the literature that project  
literature on um states and state  
emergence is written  
and this entails agriculture uh trade a  
lot of ways for basically wealth to be  
centralized and powerful people to be  
able to constrain everyone else  
awesome and that like speaks very well  
to um how  
states get a grip on their subjects and  
themselves um and what is the the common  
grounding maybe that um  
is used to get like to get a  
heterogeneous  
um  
understanding of states oh sorry  
emergency alert ah  
your attention please your attention  
please  
update in action policy requested from  
the state  
exactly exactly so so now states get a  
grip on us through like our our  
cognitive offloading into the knee  
strike  
um stephen did you want to say something  
yeah i mean this also speaks to that  
challenge of being able to get a grip on  
things which are beyond the adjacent  
scale like that you know they're trying

to get you know they have to use they  
have these technologies they can reach  
out to everyone but at the same time  
it's kind of  
they can't easily get a grip  
so there's um there's some interesting  
points there around um  
you know because there's a few scales  
between us and the state right  
um  
and that's that's interesting  
and technological development  
like  
little tiny chips that track where you  
move  
that makes it a little easier to get a  
grip on where somebody is moving  
so it's a co-evolution with the states  
and technology going back to  
agriculture and writing and ledgers and  
now we're in a new phase of ledgers from  
zero one two three  
nice stephen  
yeah and of course the challenge is that  
they get a grip on what you're doing but  
they their influence is about why you're  
doing it  
decontextualized so  
they think someone's  
doing x  
or y and  
it may be wildly different  
you know they may be underestimating the  
risk or they may be radically  
overestimating the risk of behavior  
awesome so i'm going to just flip  
through these next few slides  
so

uh dense settlements the what the  
function of the dense settlement did was  
allowed for economic codependency and we  
talked earlier about the functional  
specialization that was like you know  
necessary for this economic  
co-dependency and it also facilitated um  
the development of  
nested social communities uh um and  
daniel wrote is their scaling  
relationship here and i think we have to  
wait and address that maybe in the in  
the dot one

um

and uh moving on it says enforcing state  
control over urban systems  
mandates building acceptance into the  
population and this is like how do you  
let the state be in charge of you how do  
you grant grip to the state  
and this is  
the people must perceive competency in  
the authoritarian

regime and there's a redistribution of  
wealth like for example paying taxes  
and also symbolic legitimacy which is  
achieved through the church  
um largely or maybe maybe there's other  
ways but but was in in the in the early  
development of cities and states largely  
um religious based

so what um

there's a lot of things that that we've  
talked about so that tie into  
formation of cities and states and it's  
large-scale coordination collective  
identity

crop redistribution which i thought was



interesting like some people grow it  
some people eat it some people take it  
some people sell it  
and then the functional specialization  
specialized rules  
and the control of the money and the  
flow of the money so all these things  
together  
unify to give a state grip and allow  
this kind of top-down control  
is there any comments on that or if  
not we can move on to the next section  
just that if those are the necessary and  
sufficient features  
when there's a crisis of competency or  
distribution of wealth or symbolic  
legitimacy  
what is that forbode  
avail do you have a an answer i'm sure  
waiting and waiting in the wings  
yes uh the answer is basically uh a  
structural  
structural demographic theory which is  
central side results in clevelandics  
which basically  
states afford the possibility for the  
elites to ignore  
the social  
maintenance and just capture wealth  
and  
when they do too much it causes  
political instabilities so  
and this entails uh fluctuating um  
protecting cycles of integration and  
distribution  
in states developments and by the way  
we in the sense of france the us and  
likely

global capitalism are in a phase of  
disintegration which  
uh needs  
which entails that we have uh you don't  
need to understand how to organize  
outside or  
in spite of  
administration and to understand where  
they come from  
cool  
so we're going to move into the last  
section of the paper which is from the  
urban metabolism to  
the collective mind  
and we're going to go back to the main  
question of the article which was what  
kind of intelligence is displayed by  
city-state systems and how is it  
embodied within their structure  
and uh maybe stephen do you want to read  
this quote from the paper  
yes  
okay so answering this question in terms  
of inactive inference entails revisiting  
our accounts of the urbanization  
dynamics as the results of city-state  
systems  
actively managing to get a grip on their  
world ie  
minimize their expected variational free  
energy  
cool and there are a couple more um  
quotes from this section uh that we can  
move on to  
daniel do you want to read this first  
one here  
basic autonomy is indeed formally  
defined as operational closure or more

precisely the capability of a network of constraints to canalize faster processes like metabolic flows into maintaining and recreating itself

i like that's kind of a twist on the kantian organism concept the teleological definition the organism is like the level it makes sense at and this takes a more operational and mechanistic approach towards closure of mechanism and

uh

constraints and autopoiesis rather than merely end directedness which doesn't immediately suggest a mechanism and then

since the material and socio-cultural landscapes of urban systems canalize human activities so as to maintain itself

the process of urbanization clearly intains the constitution of cities as basic life

food comes in waste goes out city's alive

awesome and i love this picture of like the elephant is the city um sir did you want to say

i want you to something out that the first point you made is that um with the mountable and moisture definition or my

account of them is a twist on the conscience idea that organism is the level

at which it makes sense

and it's basically the movement that monteville and monsieur and moreno make

because

uh in activism at first it's surrounded  
in concertal philosophy and feminine  
energy

um

[Music]

this this is the grounding in which uh  
varela and maybe machrena i'm not sure  
articulated uh the view that cognition  
is in fact grounded in life  
and what uh

mostly people and the basque center for  
life sciences i think it's called try to  
do is to mechanize it to personalize it  
to ground it into specific uh chemical  
biochemical processes so it's you  
precisely captured what this account is  
meant to do

awesome

and good good job inferring daniel  
based on based on the um you know  
stigmatic cues left in the niche uh and  
and we can move on to a couple more  
quotes from the paper um

this is one that really resonated with  
me and it's paraphrasing jablonka and  
lam in 2006 this paper that's here the  
evolution of information in the major  
transitions

says a full-fledged evolutionary  
transition typically entails the  
emergence of new forms of information  
systems mediating regulation at the  
collective level and or the heritability  
of collective traits

and then another quote from the paper  
is also taken from this gaudian crawl  
paper in 2014

it says the emergence of city-states  
with their professional classes and  
their mediating institutions  
based on writing technologies constitute  
a clear case of such a transition  
especially in the case of the late  
agricultural civilizations and i i  
totally get this and also it really  
speaks to how stigmatic cues left in the  
niche environment can contribute to  
the formation of our collective brain i  
mean writing like we can leave stigmatic  
cues for generations yet to follow and  
some writing in stone right like so this  
is like still we can read it many many  
many many ages and generations later  
which i thought that was really uh  
poignant stephen  
yeah one thing i'd be interested in in  
terms of the state is um  
how much different  
areas of society or the state  
are trying to achieve similar types of  
outcome  
so when there's um and when there's i  
mean we've just had that case of um  
that video of  
a  
cortez being murdered and uh you know so  
like within  
the state you have  
internal conflicts you know within  
within the state itself um but also  
how much does the people at different  
scales have to be aligned or how much  
will the state itself kind of do what  
states do so to speak as its own mind  
awesome thanks daniel

yeah i agree about  
not every single action in the state  
necessarily being the same like if you  
looked at the body you said well no  
we're supposed to be making glucose why  
are there cells that are breaking it  
down or like  
there can be sections that are doing  
processes that seem directly opposite to  
each other  
but that can be part of a broader  
regulatory process so  
it  
brings into question you know what are  
the official  
processes versus the things that are in  
the gray zone of the state  
and how do we go from this model into  
more useful forms  
nice thanks  
um and so we can move now to another  
quote from the paper that is from  
paraphrasing  
this paper that we actually just did in  
a recent live stream um how to count  
biological minds by  
matthew sims  
and in  
in the thinking like a state paper it  
says if the nested hierarchy of  
regulation systems enables adaptive  
behavior at the system scale it  
effectively enacts a collective mind  
regardless of  
whether system components retain their  
cognitive ability  
and this speaks back to what we were  
saying earlier about the the top level

kind of constraining the lower levels  
like maybe there's like a uh  
once you contribute to the collective  
then you know you've contributed to  
something greater than yourself and your  
your own cognitive ability perhaps  
doesn't matter so much  
any thoughts on that or we can move on  
to some more topics from the paper  
daniel do you want to read this  
quote  
the closest analog  
of our intuitive picture of a mind in  
urban systems is their most disembodied  
mechanism for adaptive regulation  
administration  
indeed just like neuronal cognition  
it entails the top-down regulation by a  
specialized population with the help of  
a specific purpose information system  
it makes me think about those tubes that  
they used to put the physical  
packages in and send them and now  
they're special information channels but  
you better be listening when that  
channel calls  
so that's like futurama like they have  
like i mean the bureaucracy like the  
administration and i mean administration  
i love this quote from the paper i do  
think like administration is totally  
like disembodied from like reality and  
bureaucracy is like so imposing but like  
in the futurama they all have like you  
know it's like space age modern  
transmission like you could just click  
docusign on the internet and like yet  
they're using like tubes literally like

to send the things off um and it takes  
seven years to get approval or something  
ridiculous like that but it like it does  
work like that right bureaucracy is is  
kind of crazy

this next quote says the institution of  
state-sponsored legal codes affords a  
way to directly transcribe

administrative expectations into  
individual minds rule of law is

therefore a core tool by which states  
create their world through building the  
basis of their own intelligent behavior  
into their socio-cultural niche and this  
is brilliant right like just yesterday

my landlord called the codes officer um  
because my neighbors have weeds right  
like i know they're not allowed to have  
weeds it's beyond the expectation of of  
like you know our neighborhood that  
nobody has weeds and everyone knows that  
this is the expectation some people  
don't maintain their weeds and we all  
have the right and duty and obligation  
to you know impose that on our niche  
environment daniel

um so the uh

expectations the normative expectations  
a lot of laws are like you know it shall  
be within two weeks after this giving  
notice that this happens so their  
expectations and then they're called  
codes and so maybe think about  
predictive coding

and it's like we can predict that this  
technology is going to be used this way  
so i know it's not exactly what people  
mean with the neural rate codes etc with



predictive processing predictive coding

but

laws are

normative they're saying what should

happen and that helps

behave as expected you want to be on the

right side of the road driving in the

u.s so that's like an expectation and

then it becomes an intersubjective

preference that is sponsored by the

state so it's just kind of cool again

showing points of contact between

people's day-to-day life

and their intuitive explanations for how

these systems work and then it's like

yes we have an ontology that picks up

right with you

and then

goes a little further than you might

have expected to

awesome serval

yes that's exactly

the idea

knows until the externalization of

expectations the fact that they are

normative is i think

incidental accidental it's a phenomenon

to their core role which is to shape

behavior

you have a project in

legal studies i don't know how you call

that

which is called i think homogeneous

ridiculous or homogeneous or some

closely associated formulation which has

a behavioral approach to bone

and

we need to do that more

yeah sounds good

um so we just have some summary um  
quotes maybe from the paper and to kind  
of summarize the ideas that were  
presented here daniel do you want to  
read one or some of these

okay i'm just picking which one i would  
want to read i think the second one is a  
lot like what we were just talking about  
with

the state defining expectations and then  
modifying its own niche that operational  
closure to

implement those expectations lest it  
dissolve

then i i like this top quote our notion  
of intelligence is based upon active  
inference a scale-free framework  
applying to any system with individuated  
boundary states

and then it's like

even the word state is upon as avail  
pointed out before it's at being state  
and french and we talk about internal  
external blanket states boundary states  
that sounds like it's going to be at the  
boundary of the empire

and then this is what it looks like to  
apply active inference to a real system  
to go from a sort of bayesian graph  
boundary notion to being like well what  
if that boundary state

country

was

intermediating and insulating between  
these other two so i like how even the  
word state

comes into play i like that i i really

like that too and i like that like that  
that's kind of a play on um like the  
first in particular physics like  
specific particular physics but also  
like physics as related to a particle um  
and and the idea of a state actually  
something when i was proofreading this  
paper was like you know i know that  
you're talking about like city states  
but it's not  
immediately obvious to the reader that  
like you might want to say like city  
states to kind of flesh that out like  
from the get-go because it is it can be  
such a play on word right it's a  
boundary state it's a blanket state it's  
an active state it's a you know city  
state daniel maybe one other thing we  
can talk about in the dot one and two is  
states as we know them  
are  
new  
and so it's only been a few generations  
like your great great great grandparents  
that's a long time ago it's a lot of  
tacos ago but  
it's a blink of the eye  
in terms of evolution  
and in terms of deep history  
and so even pre westphalia that was not  
so long ago  
and there's so many  
world traditions and so just  
realizing that we're in a very  
non-stationary moment  
with the development and the  
implementation of states  
it's really important because it can be

easy to be like oh it's a grind all the  
states in the u.n and you know 50 states  
in the u.s it's like  
even 70 years ago there weren't that  
many states in the u.s and there was  
different countries in the world so  
things are changing it just sometimes  
it's at a different time scale which  
makes sense because they're bigger and  
slower entities than we are  
cheval

i want to point out two things first the  
paw is not intended but no i will claim  
it is and no one will check  
um the second is that  
the received you is that  
states emerge with agriculture it is  
what gordy and kroll said in the paper i  
cited  
um it is not accurate or we don't know  
whether it separates uh you have a  
pretty good single review of the  
literature of the  
state emergence and the fact is we do  
not know what human societies were like  
before written history we do not know  
so i  
and most importantly it's um  
something uh grabber and one grow to a  
leading anthropologist in historical  
apology like to uh rave about  
uh there are kinds of conversations that  
are authoritarian there are kinds of  
transactions that are not so even with  
if we are  
signs of i don't know great ritual  
states or anything we don't know whether  
those kind of organization

were like what we know today because uh  
western national states are extreme in  
kind of extractive and corrective  
behavior but they are shaped by war  
within europe and between europe and um  
colonial colonialized uh countries  
uh so i prefer i much prefer to ground  
uh the history in which i talk uh in  
europe and history in a modern history  
after westfalia after the birth or  
in the roots of the birth of the  
national states because  
it's pretty hard  
to talk about the roots of sumerian  
states we need to do so because  
those are states we need to do so if we  
want to talk about what states are but  
if you want to talk about what our  
states are we need to go uh in west  
faria we didn't mean not to go in summer  
or egypt or whatever  
so just to kind of play off of that um  
extractive and corrosive behavior  
and that states are often born out of  
war um something that you mention in the  
paper but um i i don't think maybe gets  
enough um  
attention is the concept of shared  
identity because in this shared identity  
like that's what makes us faction off  
into our own state we're going to divide  
and become something new  
and daniel was saying how we have so  
many states now so many more states than  
we've ever had before and this like  
factionalism and we all have like this  
distinct identity is this going to breed  
more war or is every state going to

become so small because they'll just  
faction and keep dividing i mean you can  
get something the most homogenous human  
culture you can imagine and they will  
find something to divide them so so is  
this good is it bad if it we keep  
dividing is it going to be like a  
homogeneous like bunch of sub-factions  
or is it just going to lead to like more  
division and more war

um

if i discuss that it will take a lot of  
time

i understood this session was excellent

we are much off track

definitely

we'll save it for the dot one

um

okay so moving on uh the collective mind  
does not directly emerge from  
inter-individual interactions but from  
slower evolving dynamical constraints  
which supervene on but do not reduce to  
human activity

therefore the proper scale of analysis  
of collective cognition is an integrated  
activity of entire urban systems and  
should more specifically be grounded in  
the broader processes by which they  
maintain and reproduce their structural  
identity understood as an operationally  
closed set of constraints underlying  
their metabolic cognitive and ecological  
activity

you know replace urban with  
brain body

just you could say the same exact thing  
the brain the body have to be an

integrated unit

finally

blue what does it mean to think like a  
state

i'm going to put your overlay over the  
video so it'll look like you're a state  
go for it

the cognitive activity of states is  
defined mechanically by the standardized  
administrative systems they use to  
understand and manipulate their world  
and functionally by their drive to build  
and enact grip over the material and  
sociocultural niche

so there it is the question the answer  
we made it through from a to z and i  
like it's not what is it like to be  
a state it's not the nagel 54 what is it  
like to be a bat

this is more of a cognitivist  
interpretation

that

helps us actually

potentially find useful

ways you know

like

i don't want to do a you know in in x  
country

you grip the state

people can kind of write their own jokes  
on that theme

nice um and we can just think about what  
maybe these

questions lead to maybe what a good  
understanding of this work might enable  
um avail mentioned earlier the  
necessity for this understanding because  
of the the kind of divide divided

space that we're in right now

what are the next steps for the fep and  
active inference

what are the goals what are you still  
curious about if you want to participate  
in either of the discussion videos get  
in touch with us we would love to have  
you on to talk about these ideas in more  
detail do you guys have any closing  
thoughts

yeah just in dot

one let's

dive in and open some threads take some  
notes

in dot two let's see where we go maybe  
somebody would like to  
pick up on one of those threads or apply  
one of these ideas and say oh i was  
working on a history of my region and so  
i wanna

see how i can apply that

so we applied active inference to get  
thinking like a state and then how do we  
apply

thinking like a state

for

something else whatever it is that  
somebody is interested in seeing  
avel thank you for coming and kind of  
guiding us through our contextualization  
video it's always a pleasure do you want  
to have the final word

i don't know and it is anything smart to  
say so

yes but

these are the words

the paper does speak for itself uh quite  
brilliantly i will say



all right authors who come on dot zero  
are the best  
no it's all good thanks for joining blue  
and facilitating see you later survival  
and thanks also to stephen so hope to  
see you all in the dot one  
with the dot two or beyond so bye  
bye  
[Music]